

The genesis of Tiger Lake

We sit down with Chris Walker to learn more about the technologies and goals pertaining to Intel's Tiger Lake

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Mithun: What were the key goals that you set out at the onset of starting on the Tiger Lake journey?

Chris: We think of Tiger Lake as a mobile-first platform. The goal with Tiger Lake, first and foremost, was to deliver breakthrough performance and not sacrifice battery life. We do this every generation and it gets harder and harder every time. It was to further enhance the integration elements and a lot of it was about advancing all the IPs that we thought would matter more to our end-users. That's why you see things like the improved imaging processor for higher quality cameras, especially now that we're having a lot more video conferences. It was about taking steps to improve our neural accelerator for low power noise suppression use cases.

The big focus was the graphics engine. If you look at what we're able to do with the Xe architecture, we can deliver the performance and media display capabilities all at the same time. In our Tiger Lake Blueprint videos, Boyd Phelps explained how the ring architecture and the super thin transistor helped better deliver that performance while not spiking power consumption and thereby enabling longer battery life. I think what we've achieved in the product is this really balanced high-performance compute and AI-creation engine while



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at the same time achieving great power characteristics and battery life.

The other goal we had with the platform was to go even deeper with our OEM partners on system designs which ultimately lead to the Evo platforms. The two went hand-in-hand. As we looked at the advancements that we could make in Tiger Lake architecturally, we then were able to match that with the second edition of the Intel Athena specifications and really tune that and see it come to life in our OEM's systems. From that standpoint, we achieved our goals and hit the mark. Some of these ideas started some three-plus years ago while some started eighteen months ago and the last six months have taught us that certain features became much more important.

Mithun: One of the things you've touched upon is battery life and we're hearing a lot of chatter around new battery tech which is going to improve

battery life by magnitudes. When such a technology does come into play, is the whole focus around optimising for power efficiency going to remain as important as it is today?

Chris: The two things that we talk about battery or energy efficiency is that even with better densities or battery cells, there's another element, that of power management. A big part of the efficiency is that you're not only focused on battery life and cooling but it's also about performance while you're on battery. So if you look at our run time charts then we provide the same level of performance whether you're plugged in or if you're on battery. So one of the goals of efficiency, tuning and those types of optimisations, is to ensure that you have a consistent experience on a laptop. They're not just desktops with screens, you expect to use it on your couch and use it when you're in your backyard or sitting in a coffee shop and still have it be just as responsive. So even though battery technology enables better densities and longer battery life, the focus on efficiency and tuning is still there to provide the overall experience.

Mithun: With regards to the Evo certification a.k.a. Athena 2.0 you've gotten a bit more descriptive regarding the criteria for the certification. So when your OEM partners do end up meeting these stringent criteria, how can these OEMs distinguish themselves from one another? Why would I pick, let's say, a Dell unit over an MSI unit if they're both Athena 2.0 certified?

Chris: The Evo or Athena 2.0 specifications aren't an exact checklist. Within the specifications we allow for ranges, not on things like the responsiveness or the battery life on FullHD or the charging or the ports. You'll see some models with FullHD while there will be others with UHD screens. We have a range of screen-to-body ratios and some OEMs will go even further. For example, Samsung made a 5G version of an Evo platform to differentiate themselves. So there are multiple technologies under the gamut of the Evo platform. So when you work with Project Athena there's a list of recommended or optional or advanced technologies that OEMs can implement.